Assignment - 3 (COA)

* Section - A

Ans: 1 Clocking methodology is the mechanism that synchronize all processor operations using a clock signal.
Every action - data transfer, computation - happens at specific clock edges.
It ensures all components like registers and ALU work in coordination without conflicts.

Ans: 2 A clock cycle is the smallest unit of time in a processor, defined by one complete tick of the clock signal. It determines how ~~much~~ fast instructions can execute - shorter cycles means higher processor speed.

Ans: 3 ~~At~~ Amdahl's Law states that overall speedup of a system is limited by the portion that cannot be parallelized. Even if part of a program is highly optimized, the serial portion restricts total performance gain.

$$\text{Speedup} = \frac{1}{(1-P) + \frac{P}{S}}$$

Ans: 4 The data path is the path of the processor that actually performs computations and data moment. It includes components like ALU, registers, buses and multiplexer working together to process instructions.

Ans: 5 The control unit or like, the commandery officer - it generates control signals that direct all operations inside the

the processor. It tells the ALU, memory, and registers what actions to perform and when.

Ans:6 In a single-cycle processor, each instruction is executed within one cycle. The design is simple but inefficient because the clock cycle must be long enough to accomodate the slowest instruction.

Ans:7 A multi-cycle processor break instruction into multiple steps across several clock cycle. This improves efficiency since each step can use a shorter clock cycle and hardware is reused.

Ans:8 Pipelining allows multiple instructions to be processed simultaneously by dividing execution into stages. While one instruction is being executed, others are decoded and fetched - like an assembly line.

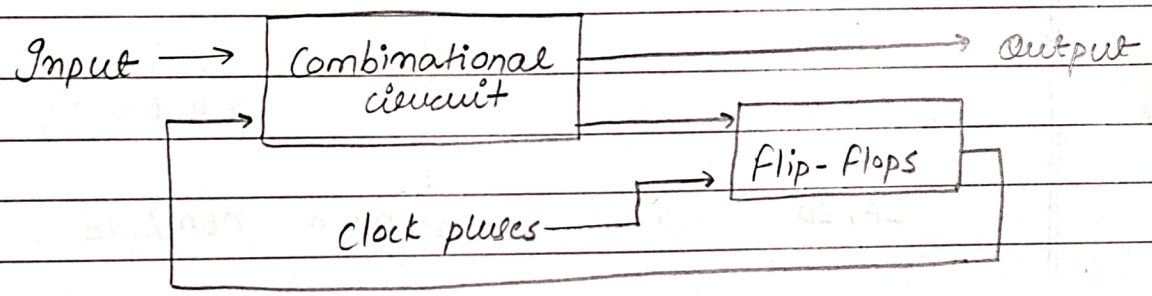
Ans:10 A data hazard occurs when an instruction depends on the result of a previous instruction that hasn't finished yet. This can lead to incorrect results if not handled properly.

Ans:11 Control hazard arise due to branch instructions that change the flow of execution. The processor may fetch wrong instructions before knowing the correct branch outcome.

Ans:12 Pipeline hazards are handled using techniques like forwarding (bypassing data), inserting stalls (delays), branch predictions, and instructions reordering. These methods ensure smooth and correct execution.

* Section - B

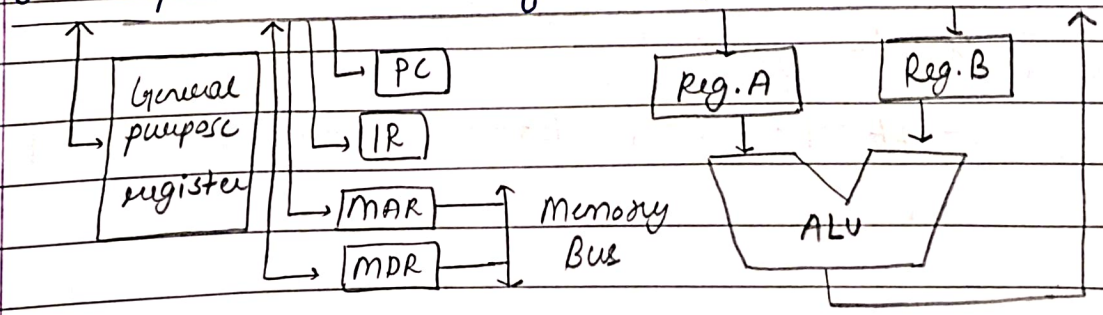
1. A clocking methodology uses a global clock signal to synchronize all registers. On each clock edges, data is transferred between registers in a controlled manner. This ensures predictable timing and avoids race conditions. Flip-Flop store data and update only at clock transitions



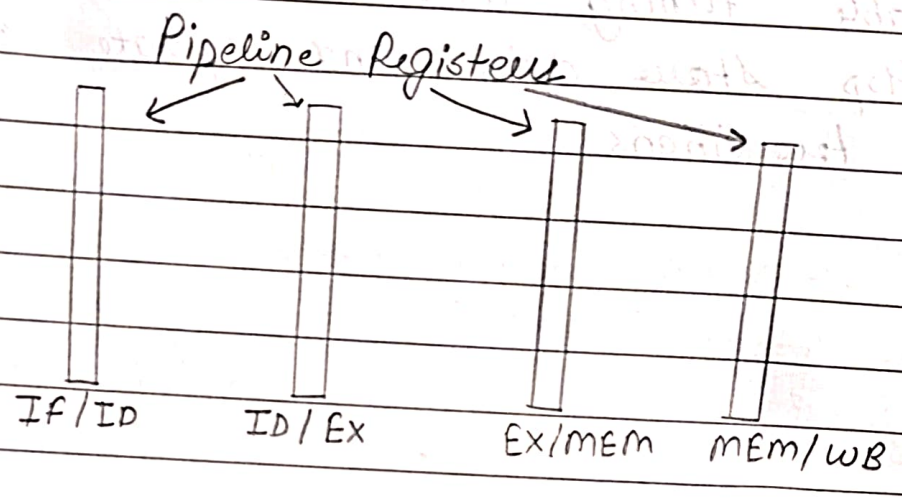
ⓐ Block diagram

ⓑ Timing diagram of clock pulses

2. The datapath includes registers, ALU, and memory connected by buses. Data moves from registers to ALU for computation and then to memory or back. & Control sigls guide operations like read, write, and execute.
It represents the flow of data inside the process



3. Instruction pipelining divides execution into sta IF, ID, EX, MEM, WB. Each stage performs pa of the instruction simultaneously. Multiple instruction are processed in parallel, improv throughput This increases efficiency but ma cause hazards.



4. Single cycle vs. Multi-cycle execution.

1) Single-cycle executes an instruction in one long clock cycle.
2) Multi-cycle divides execution into smaller steps over multiple cycles.
3) Multi-cycle is more efficient and reduces clock time.
4) Single-cycle is simple but slower due to to longer cycle duration.

Single-cycle:

|- - - - - - Instruction - - - - - - |

Multi-cycle:

|--IF--|--ID--|--EX--|--MEM--|--WB--|

* Section - C

1. Frequent data dependencies cause data hazards, when an instruction needs result not yet computed. This leads to incorrect or delays. It can be solved using data forwarding (bypassing) or pipeline stalls. Forwarding reduces delay by directly passing result between stages.

2. $$S = \frac{1}{(1-P) + \frac{P}{S_P}}$$

Here, $P = 0.8$, $S_P = 4$

Speedup $= 1 / [(0.2) + (0.8/4)] = 1 / (0.2 + 0.2) = 1/0.4$
$= 2.5$

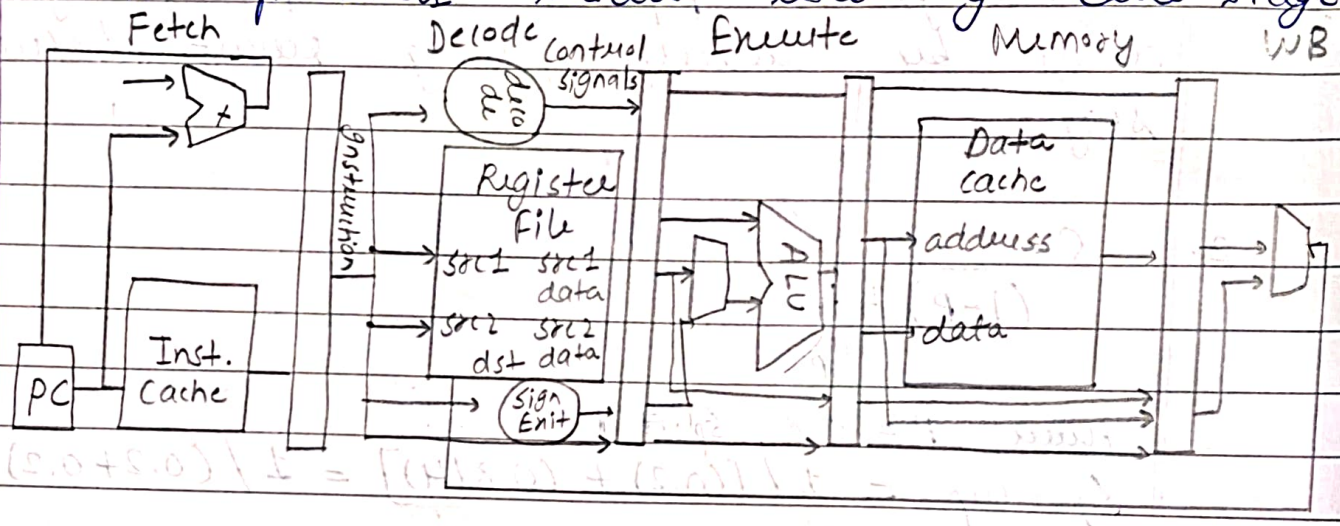
So, overall speedup = 2.5 times

3. Multi-cycle processors use shorter clock cycles by breaking into steps. Each instruction uses only required hardware, improving utilization. This reduces wasted time compared to single cycle. Hence, multi cycle is more efficient.

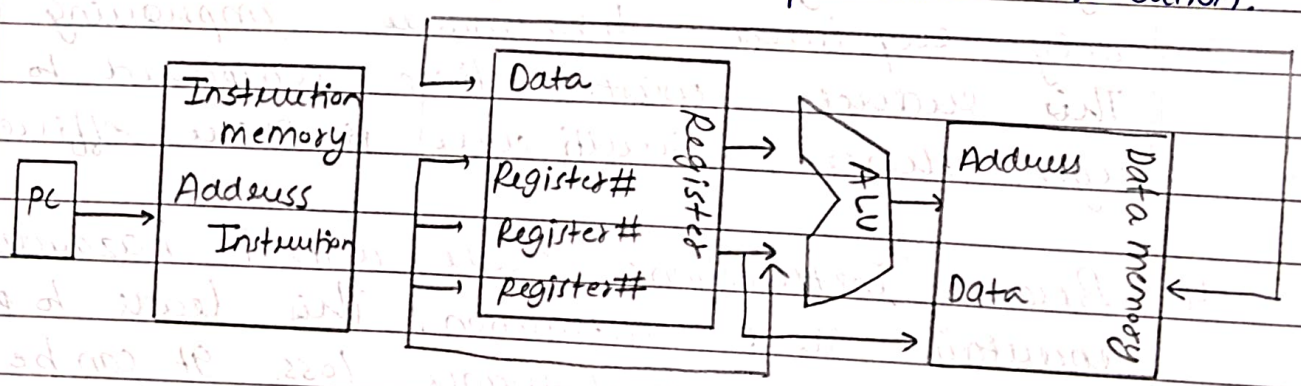
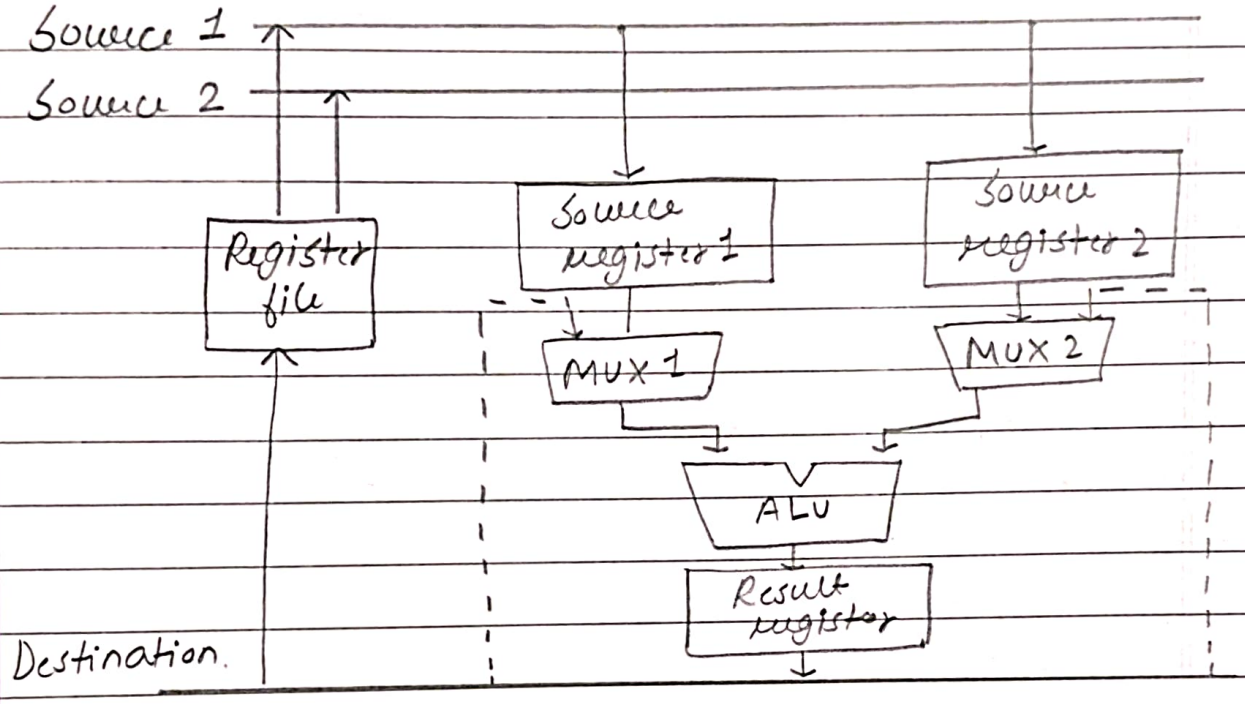
4. Branch instructions cause control hazards due to uncertain next instruction. This leads to pipeline flushing and performance loss. It can be reduced using branch prediction techique. Better prediction minimizes stalls and improves efficiency.

* Section- B

1. The Execute (EX) stage performs arithmetic/logic operations using the ALU.
It also calculates memory addresses for load / store instructions. This stage is critical as it produces results used by later stages.



2. Using a shared ALU reduces hardware cost and complexity. Instead of multiple ALUs, one unit performs all operations. This saves space but may slightly limit parallel execution.



3. This is a RAW data hazard.
It can be solved using data forwarding, sending result directly to next stage.
If not possible, introduce a pipeline stall.